

CO4 – AT3

Course Code: MLA0201

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Course Name: Fundamentals of Machine Learning

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Comparative Analysis

Question 1: Bayesian Networks vs Markov Random Fields for Disease Diagnosis

A healthcare diagnosis system must represent uncertain relationships between diseases, symptoms, and medical history.

Parameter	Bayesian Network (BN)	Markov Random Field (MRF)
Graph Structure	Directed graph	Undirected graph
Dependencies	Represents cause-and-effect relationships	Represents mutual relationships between variables
Conditional Independence	Defined using parent-child relationships	Defined using neighbouring nodes
Inference	Uses CPTs and probabilistic inference	Uses potential functions and joint probability
Uncertainty Handling	Explicitly models conditional probabilities	Models uncertainty through joint relationships
Computational Complexity	Efficient for small structured networks	Can become computationally expensive
Healthcare Use	Suitable for symptoms causing disease predictions	Useful for complex interacting relationships

Critical Evaluation

Bayesian Networks are particularly suitable for healthcare diagnosis because medical relationships can often be represented directionally. For example:

Medical History → Disease → Symptoms

This structure makes the model easier to interpret and supports inference when some patient information is missing.

MRFs are useful when relationships are symmetric, but disease diagnosis usually requires reasoning about conditional and causal relationships.

Conclusion

Bayesian Networks are more appropriate for the healthcare diagnosis system. They provide clear dependency relationships, effective uncertainty handling, and interpretable probabilistic predictions.

Question 2: Hidden Markov Models vs Conditional Random Fields for Activity Recognition

An autonomous vehicle must recognize driving activities from continuous sensor observations over time.

Parameter	Hidden Markov Model (HMM)	Conditional Random Field (CRF)
State Representation	Uses hidden states	Predicts labels for sequences
Observation Modeling	Models observation probability	Uses multiple input features
Learning Approach	Generative learning	Discriminative learning
Contextual Features	Limited feature flexibility	Handles many contextual features
Prediction Accuracy	Effective for simple sequential data	Often better with rich features
Computational Efficiency	Generally efficient	Training can be more expensive
Scalability	Suitable for simpler models	Can handle complex feature sets

Critical Evaluation

HMMs model how hidden states change over time and are computationally efficient. However, they make stronger assumptions about observations.

CRFs can use multiple sensor features simultaneously, such as:

- Speed
- Steering angle
- Acceleration
- Brake pressure
- Camera or sensor information

This makes CRFs more suitable when driving activity depends on rich contextual information.

Conclusion

CRF is recommended for autonomous vehicle activity recognition because it can incorporate multiple contextual sensor features and model dependencies between consecutive activity labels. HMM can still be preferred when computational simplicity is the main requirement.

Question 3: Bayesian Networks vs Markov Random Fields for Financial Risk Prediction

A financial analytics system must analyze customer transactions and historical financial information to predict risk.

Parameter	Bayesian Network (BN)	Markov Random Field (MRF)
Graph Representation	Directed graphical model	Undirected graphical model
Dependency Modeling	Parent-child conditional dependencies	Mutual dependencies between variables
Inference Techniques	Bayesian inference using CPTs	Inference using potential functions

Parameter	Bayesian Network (BN)	Markov Random Field (MRF)
Learning	Structure and parameter learning	Parameter and structure estimation
Generalization	Handles missing information effectively	Captures complex variable interactions
Computational Requirements	Usually easier to interpret	Can require expensive inference
Financial Suitability	Strong for probabilistic risk prediction	Useful for interconnected relationships

Critical Evaluation

Financial risk prediction often involves directional relationships. For example:

Income → Financial Behaviour → Credit Risk

and

Transaction History → Risk Prediction

Bayesian Networks can represent these dependencies clearly and estimate probabilities even when some customer information is unavailable.

MRFs are useful when financial variables have strong mutual interactions, but their undirected structure can make interpretation more difficult.

Conclusion

Bayesian Networks are more effective for this financial risk prediction application. Their directed structure, probabilistic inference capability, and ability to handle uncertainty and missing information make them practical for financial analytics.

Overall Comparative Insight

Application	Compared Models	Recommended Model
Healthcare Disease Diagnosis	BN vs MRF	Bayesian Network
Autonomous Vehicle Activity Recognition	HMM vs CRF	Conditional Random Field
Financial Risk Prediction	BN vs MRF	Bayesian Network

The model selection depends on the structure of the problem: **Bayesian Networks are strong for directional probabilistic relationships, CRFs are effective for feature-rich sequence labeling, and MRFs are useful for symmetric dependencies.**